CHAPTER 7

BINOMIAL THEOREM

CHAPTER ESSENCE AT A GLANCE:
Expanding high powers like $(x + y)^{10}$ without multiplying 10 brackets by hand! Pascal's triangle gives the symmetric coefficients, powers of the first term decrease from n to 0, and powers of the second term increase from 0 to n. The general term formula unlocks any single term in 10 seconds!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Binomial Theorem Statement:** $(a + b)^n = {}^nC_0\, a^n + {}^nC_1\, a^{n-1}\, b + {}^nC_2\, a^{n-2}\, b^2 + \ldots + {}^nC_n\, b^n = \Sigma\, {}^nC_r\, a^{n-r}\, b^r$. Total terms = n + 1.
- **General (r + 1)th Term:** $T(r+1) = {}^nC_r\, a^{n-r}\, b^r$. Remember: 5th term has r = 4; (r + 1) index is 1 greater than combination index r!
- **Middle Term Rule:** If n is even, there is ONE middle term: T (n over 2 + 1). If n is odd, there are TWO middle terms: T ((n+1) over 2) and T ((n+3) over 2).
- **Term Independent of x:** Write general term T (r+1), collect all powers of x into x^(expression), equate exponent to 0, and solve for integer r.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$(a+b)^n = \sum_{r=0}^{n} {}^nC_r a^{n-r} b^r$$

Standard Binomial Theorem for Positive Integral Exponents

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

General Term in the Expansion of $(a + b)^n$

$$\sum_{r=0}^{n} {}^nC_r = 2^n \quad \text{and} \quad \sum_{r=0}^{n} (-1)^r {}^nC_r = 0$$

Sum of Binomial Coefficients and Alternating Series

$$(1+x)^n \approx 1 + nx \quad (|x| \ll 1)$$

First-Order Linear Binomial Approximation

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** The (r + 1)th term has index r in the combination! E.g. To find the 5th term, use r = 4, NOT r = 5.
- **Alert:** To find the term independent of x in an expansion, set the total exponent of x in the general term equal to 0 and solve for r!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Expand using Binomial Theorem: $(1 - 2x)^5$. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Expansion]** $(1 - 2x)^5 = 5C0\ (1)^5 - 5C1\ (1)^4\ (2x) + 5C2\ (1)^3\ (2x)^2 - 5C3\ (1)^2\ (2x)^3 + 5C4\ (1)(2x)^4 - 5C5\ (2x)^5$. [1 Mark]

• **[Step 2: Coefficients]** $= 1 - 5(2x) + 10(4x^2) - 10(8x^3) + 5(16x^4) - 32x^5 = 1 - 10x + 40x^2 - 80x^3 + 80x^4 - 32x^5$. [1 Mark]

• **[Boxed Result: $1 - 10x + 40x^2 - 80x^3 + 80x^4 - 32x^5$]**

2. Expand using Binomial Theorem: $(x \text{ over } 3 + 1 \text{ over } x)^5$. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Expansion]** $= 5C0\ (x \text{ over } 3)^5 + 5C1\ (x \text{ over } 3)^4\ (1 \text{ over } x) + 5C2\ (x \text{ over } 3)^3\ (1 \text{ over } x)^2 + 5C3\ (x \text{ over } 3)^2\ (1 \text{ over } x)^3 + 5C4\ (x \text{ over } 3)(1 \text{ over } x)^4 + 5C5\ (1 \text{ over } x)^5$. [1 Mark]

• **[Step 2: Simplification]** $= x^5 \text{ over } 243 + 5\ (x^3) \text{ over } 81 + 10\ x \text{ over } 27 + 10 \text{ over } (9\ x) + 5 \text{ over } (3\ x^3) + 1 \text{ over } x^5$. [1 Mark]

• **[Boxed Result: x^5 over 243 + $5x^3$ over 81 + 10x over 27 + 10 over (9x) + 5 over ($3x^3$) + 1 over x^5]**

3. Using Binomial Theorem, evaluate: $(99)^5$. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Binomial Setup]** $99^5 = (100 - 1)^5 = 5C0\ (100)^5 - 5C1\ (100)^4 + 5C2\ (100)^3 - 5C3\ (100)^2 + 5C4\ (100) - 5C5\ (1)$. [1 Mark]

• **[Step 2: Powers of 100]** $= 10{,}000{,}000{,}000 - 5(100{,}000{,}000) + 10(1{,}000{,}000) - 10(10{,}000) + 5(100) - 1 = 10{,}000{,}000{,}000 - 500{,}000{,}000 + 10{,}000{,}000 - 100{,}000 + 500 - 1$. [1 Mark]

• **[Step 3: Final Arithmetic]** Positive terms = 10,010,000,500. Negative terms = 500,100,001. Difference = 9,509,900,499. [1 Mark]

• **[Boxed Result: $(99)^5 = 9{,}509{,}900{,}499$]**

4. Using Binomial Theorem, evaluate: $(102)^4$. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Binomial Setup]** $(102)^4 = (100 + 2)^4 = 4C0\ (100)^4 + 4C1\ (100)^3\ (2) + 4C2\ (100)^2\ (2^2) + 4C3\ (100)(2^3) + 4C4\ (2^4)$. [1.5 Marks]

• **[Step 2: Calculation]** $= 100{,}000{,}000 + 4(1{,}000{,}000)(2) + 6(10{,}000)(4) + 4(100)(8) + 16 = 100{,}000{,}000 + 8{,}000{,}000 + 240{,}000 + 3{,}200 + 16 = 108{,}243{,}216$. [1.5 Marks]

• **[Boxed Result: $(102)^4 = 108{,}243{,}216$]**

5. Using Binomial Theorem, prove that $6^n - 5n$ always leaves remainder 1 when divided by 25 for all positive integers n. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Express 6 as (1 + 5)]** $6^n = (1 + 5)^n = nC0 + nC1\ (5) + nC2\ (5^2) + nC3\ (5^3) + ... + nCn\ (5^n)$. [1 Mark]

• **[Step 2: Expansion Terms]** $6^n = 1 + 5n + 25$ [nC2 + nC3 (5) + ... + nCn (5^(n - 2))]. [1 Mark]

• **[Step 3: Rearranging]** $6^n - 5n = 1 + 25\ k$, where k is a positive integer. This shows that $6^n - 5n$ divided by 25 gives quotient k and remainder 1. Hence proved. [1 Mark]

• **[Boxed Result: $6^n - 5n = 25\ k + 1$ (Remainder is always 1)]**

6. Find an approximation of $(0.99)^5$ using the first three terms of its binomial expansion. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Setup]** $(0.99)^5 = (1 - 0.01)^5$. [1 Mark]

• **[Step 2: First Three Terms]** $= 5C0\ (1)^5 - 5C1\ (1)^4\ (0.01) + 5C2\ (1)^3\ (0.01)^2 - ... = 1 - 5(0.01) + 10(0.0001) - ...$ [1.5 Marks]

• **[Step 3: Calculation]** $= 1 - 0.05 + 0.0010 = 0.9510$. [1.5 Marks]

• **[Boxed Result: $(0.99)^5 \approx 0.9510$]**

7. Find the term independent of x in the expansion of: $(3x^2 \text{ over } 2 - 1 \text{ over } (3x))^9$. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: General Term]** T (r + 1) = 9Cr $(3x^2 \text{ over } 2)^{(9 - r)}$ $(-1 \text{ over } (3x))^r$ = 9Cr $(3 \text{ over } 2)^{(9 - r)}$ $(-1 \text{ over } 3)^r$ $(x^2)^{(9 - r)}$ $(x^{-1})^r$ = 9Cr $(3 \text{ over } 2)^{(9 - r)}$ $(-1 \text{ over } 3)^r$ $x^{(18 - 2r - r)}$ = 9Cr $(3 \text{ over } 2)^{(9 - r)}$ $(-1 \text{ over } 3)^r$ $x^{(18 - 3r)}$. [2 Marks]

• **[Step 2: Independent of x Condition]** Exponent of x must be zero: 18 - 3r = 0 ⇒ 3r = 18 ⇒ r = 6. [1 Mark]

• **[Step 3: Numerical Value]** T_7 = 9C6 $(3 \text{ over } 2)^{(9 - 6)}$ $(-1 \text{ over } 3)^6$ = 9C3 $(3 \text{ over } 2)^3$ $(1 \text{ over } 3^6)$ = [(9 × 8 × 7) over 6] x (27 over 8) x (1 over 729) = 84 x (27 over 8) x (1 over 729) = (84 over 8) x (27 over 729) = (21 over 2) x (1 over 27) = 21 over 54 = 7 over 18. [1 Mark]

• **[Boxed Result: Term independent of x = 7 over 18]**